UV & marimo

The new Python tools you should know about...

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"RDVS Numériques" du département DuMAS de l'I2M
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Summary

- Why use a **PVE** (Python Virtual Environment) for your Python projects?
- **UV**: An extremely fast Python package and project manager, written in Rust
- **marimo**: Python notebooks to replace Jupyter notebooks?

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- **UV**: An extremely fast Python package and project manager, written in Rust
- **marimo**: Python notebooks to replace Jupyter notebooks?

Ressources

- GitHub repository for the PDF slides and the notebooks: github.com/cjlux/marimo_notebooks

Who am I

Season 1

- Industrial career of ~10 years
  Signal Processing at Thomson-SINTRA
  OO design & C++ development at IT companies (Capgemini...)
- I joined the I2M lab of ENSAM (Bordeaux) as lecturer
  I led the design and development of the GranOO workbench (*DEM computing workbench*)
  I taught Scientific & OO programming in Python at ENSAM
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Season 2

- Since Sept. 2023 $\rightsquigarrow$ still working on *AI / Data processing* with Python as a freelancer.

Python Virtual Environment (PVE)

Interest of a PVE

State of the art in Python programming:
$\rightsquigarrow$ developing each project within a **PVE** providing an **isolated**, **dedicated**, **persistent** environment:

- containing a specific installation of Python
- **independent** of other Python installations likely to coexist on the same machine
- **independent** of computer updates.

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Python Virtual Environment (PVE)

A simple implementation...

A PVE is based on a **dedicated disk tree** that houses:

- the desired version of the Python interpreter
- the specific versions of the Python modules for the project
- the specific associated tools (binaries, scripts...).

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Easy to use...

You can **create**, **duplicate**, **delete** and **recreate** a PVE easily, without impacting other Python installations possibly present on your computer.

Python Virtual Environment (PVE)

Commonly used tools to manage Python project within a PVE

- the **conda** command, available if you have installed miniconda (or Anaconda) on your computer
- the **venv** Python module [docs.python.org/3/library/venv.html]
- the **poetry** tool [poetry.org]

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The new way...

- Recently a new tool has been unveiled to manage Python project within a PVE : **uv**
- It is considerably **faster** and **easier** to use...

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UV ⇝ the new Python project/PVE manager

😀 UV from Astral

docs.astral.sh/uv/

- For more than a decade, **pip** has served as the cornerstone of Python's packaging
- A lots of tools for Python modules / PVE manager: **poetry**, **virtualenv**, **venv**, **conda**...
- By 2025, **uv** became **pip** replacement:
  - ⇝ unified environment, dependency, and tool manager
  - ⇝ capable of handling every stage of the development lifecycle from interpreter installation to dependency synchronization and script execution.

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Installing UV

- **macOS & GNU/Linux**

  In a terminal window:

  ```
  curl -LsSf https://astral.sh/uv/install.sh | sh
  ```

- **Windows**

  In a PowerShell window:

  ```
  powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
  ```

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Command: `uv init [dir]`

- Create a project directory with a PVE inside:

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Creates/uses the directory `myproj` and creates inside:

- `.gitignore` & `README.md` files and a ready to use `.git` sub-dir
- `pyproject.toml`: lists the version of the installed Python modules
- `.python-version`: the version of the PVE's interpreter
- `uv.lock`: detailed informations about modules dependencies
- as soon as necessary creates the `.venv` sub-directory to hold the PVE with the right version of the Python interpreter.

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UV

Terminal

Terminal Terminal

```
jlc@kong:tmp$ uv init myproj
Initialized project `myproj` at `/home/jlc/tmp/myproj`
jlc@kong:tmp$ cd myproj
jlc@kong:myproj$ tree -a -L 1
.
├── .git
├── .gitignore
├── main.py
├── pyproject.toml
├── .python-version
└── README.md

2 directories, 5 files
jlc@kong:myproj$ cat .python-version
3.13
jlc@kong:myproj$ uv run python
Using CPython 3.13.12
Creating virtual environment at: .venv
Python 3.13.12 (main, Feb 12 2026, 00:45:41) [Clang 21.1.4 ] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
jlc@kong:myproj$
```

PVE with the right version of the Python interpreter.

UV ⇝ the new Python project/PVE manager

Command: `uv init [dir] --python` ⇝ `uv init [dir] -p`

- To choose the version of the PVE's Python interpreter
- Create a project directory with a PVE inside:

```
uv init myproj -p 3.x
```

- Initialize a PVE within an existing directory:

```
cd myproj
uv init -p 3.x
```

UV ⇝ the new Python project/PVE manager

Command: `uv init [dir] --python` ⇝ `uv init [dir] -p`

- To choose the version of the PVE's Python interpreter
- Create a project directory with a PVE inside:

```
uv init myproj -p 3.x
```

- Initialize a PVE within an existing directory:

```
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uv init -p 3.x
```

- Creates/uses the directory `myproj` with all the stuff inside...
- As soon as necessary creates the `.venv` sub-directory to hold the PVE with the the desired version of the Python interpreter.

UV

Terminal

Terminal | Terminal

```
jlc@kong:tmp$ uv init myproj -p 3.15
Initialized project `myproj` at `/home/jlc/tmp/myproj`
jlc@kong:tmp$ cd myproj
jlc@kong:myproj$ tree -a -L 1
.
├── .git
├── .gitignore
├── main.py
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├── .python-version
└── README.md

2 directories, 5 files
jlc@kong:myproj$ cat .python-version
3.15
jlc@kong:myproj$ uv run python
Using CPython 3.15.0a6
Creating virtual environment at: .venv
Python 3.15.0a6 (main, Feb 12 2026, 00:43:06) [Clang 21.1.4 ] on linux
Type "help", "copyright", "credits" or "license" for more information.
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jlc@kong:myproj$
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UV ⇝ the new Python project/PVE manager

Command: `uv add`

- To add a Python module to the PVE
- Let **uv** choose the last version of the module compatible with the interpreter version:

```
uv add module_name
```

- Choose manually a version of the module:

```
uv add module_name==version_number     # excatly this version
uv add module_name>=version_number     # at lest this version
uv add module_name>=n1,<n2             # between n1 and n2
```

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- **Very fast** (parallel) download of the module and its dependencies
- **Very efficient cache mechanism**
- Installs the module in the `.venv` directory
- Updates the file `pyproject.toml`
- Creates/updates `uv.locks` (exact version of the installed modules).

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Command: `uv remove`

- To remove a Python module to the PVE:

```
uv remove module_name
```

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Command: `uv remove`

- To remove a Python module to the PVE:

```
uv remove module_name
```

- Removes the module from the `.venv` directory
- Updates the files `pyproject.toml` and `uv.locks`.

UV ⇝ the new Python project/PVE manager

Command: `uv pip list`

- To list the Python modules loaded in the PVE

```
uv pip list
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UV ⇝ the new Python project/PVE manager

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- To list the Python modules loaded in the PVE

```
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```

- lists the modules that have been installed in `.venv` directory

UV ⇝ the new Python project/PVE manager

Command: `uv run`

- To run a Python program within the PVE:

```
uv run python_file.py
```

- To run a command:

```
uv run command
```

- Tries to run the file `python_file.py` using the Python modules and interpreter from the PVE
- Tries to run a file `command.py` within the project PVE or others environments...

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- Tries to run a file `command.py` within the project PVE or others environments...
  Example: `uv run jupyter lab`

UV ⇝ the new Python project/PVE manager

Use case 1: Try some Python code within a PVE..

- Let say you found a tuto on Internet with a sample Python code like this one:
  CustomTkinter - A Complete Tutorial...
- you want to try the Python code **without breaking your system's or session's Python installation**...
- the **uv** solution:
  - ⇝ create a PVE inside a directory: `uv init uv-customTk`
  - ⇝ go to the created directory: `cd uv-customTk`
  - ⇝ add modules to the PVE: `uv add customtkinter`
  - ⇝ run the Python code within the PVE: `uv run main.py`
  - ⇝ play with the code... when finished, delete the directory or keep it.

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Getting Started

Installation

Before diving into CustomTkinter, you need to install it. CustomTkinter is available on PyPI and can be installed using pip:

```
pip install customtkinter
```

First Application

Let's create a simple "Hello World" application to get started:

```
import customtkinter as ctk

# Set appearance mode and default color theme
ctk.set_appearance_mode("dark")
ctk.set_default_color_theme("blue")

# Create the main window
app = ctk.CTk()
app.title("Hello CustomTkinter")
app.geometry("400x200")

# Add a label
label = ctk.CTkLabel(app, text="Hello, CustomTkinter!", font=("Helvetica"
label.pack(pady=20)

# Add a button
button = ctk.CTkButton(app, text="Click Me", command=lambda: print("Butto
button.pack(pady=10)

# Run the application
app.mainloop()
```

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Use case 1: Try some Python code within a PVE..

- Let say you found a tuto on Internet with a sample Python code like this one:
  CustomTkinter - A Complete Tutorial...
- you want to try the Python code **without breaking your system's or session's Python installation**...
- the **uv** solution:
  - ⇝ create a PVE inside a directory: `uv init uv-customTk`
  - ⇝ go to the created directory: `cd uv-customTk`
  - ⇝ add modules to the PVE: `uv add customtkinter`
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Use case 2: Initialize a new Python Project within a PVE..

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- Many Git Repositories use **uv** to manage their Python project
- For example: UV-Marimo_at_I2M
- Clone the repository (`git clone ....`)
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- Go to the root directoey of the repository and synchronize the PVE to download and install all the required Python modules:

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- Read the REDAME...

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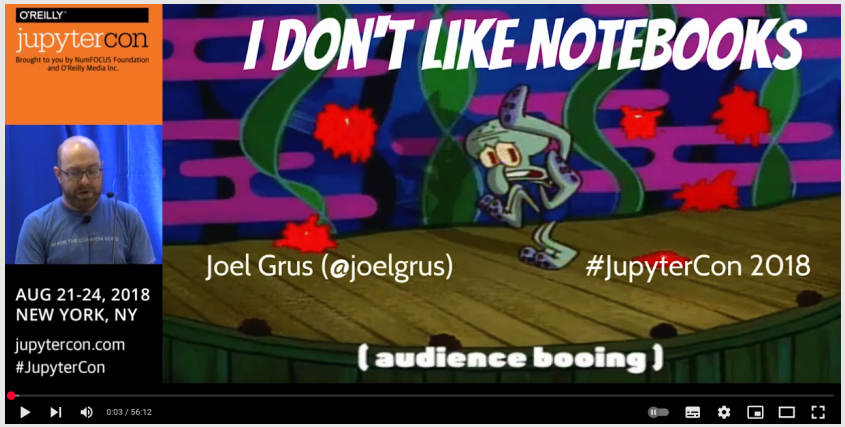
some people don't like working with Jupyter notebooks... [**Grus1**]



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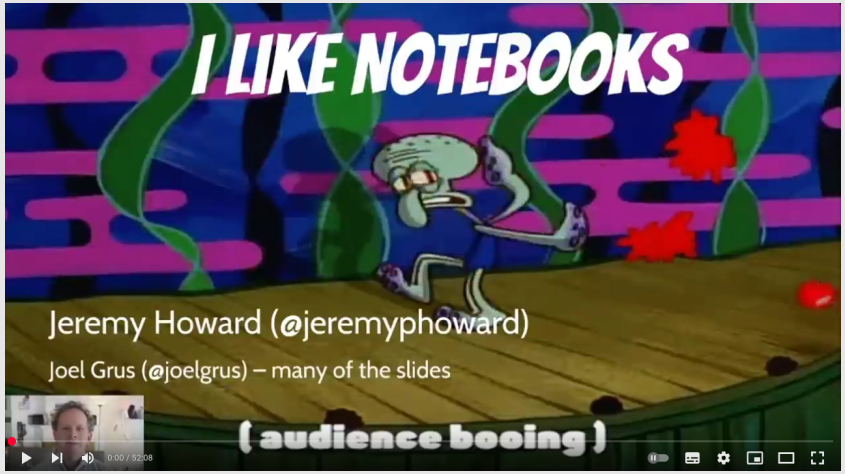

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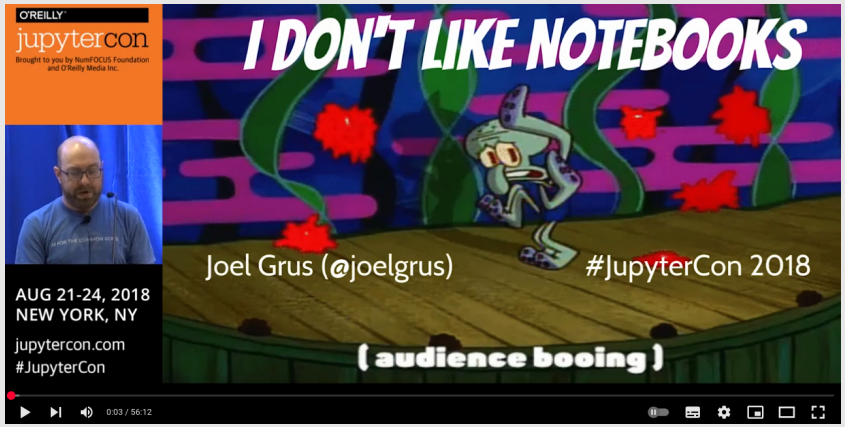

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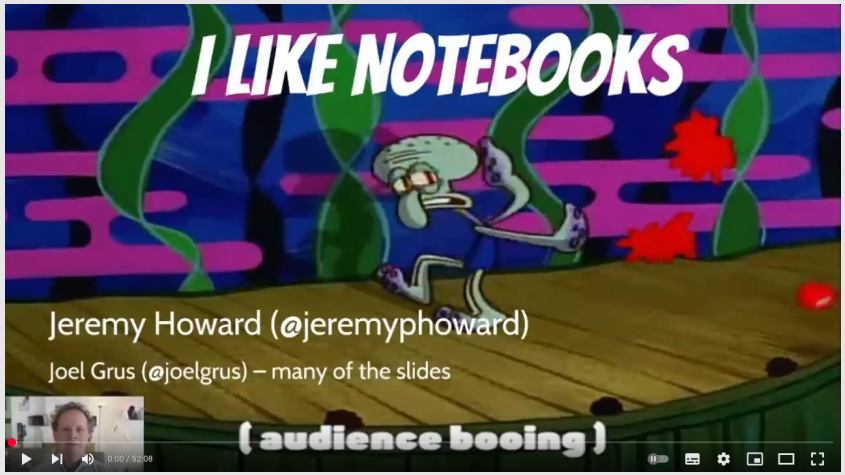

🤔 What are the main *pros/cons* of Jupyter Notebooks...

Jupyter notebooks *pros/cons*

Some advantages

- Very well known...
- A very large number of published notebooks, covering very varied application areas
- Huge user community, a lot of tutos
- Powerfull to create documentation with embeded graphs, pictures...
- Nicely permissive: interactive asynchronous execution of cells
- and many more...

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- **Lack of reproducibility** the code you see doesn't necessarily match the outputs on the page or the program state.
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Marimo notebooks: pure Python

docs.marimo.io

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Notebooks/simple.py

Simple marimo notebook to show the content of the Python file

```
import numpy as np
```

```
x = np.arange(1,5)
```

```
array([1, 2, 3, 4])
```

```
x
```

```
import marimo as mo
```

PYTHON | MARKDOWN | SQL | GENERATE WITH AI

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PYTHON MARKDOWN

```
import marimo

__generated_with = "0.13.15"
app = marimo.App(width="medium")


@app.cell(hide_code=True)
def _(mo):
    mo.md(r"""Simple marimo notebook to show the content of the Python file
    return


@app.cell
def _():
    import numpy as np
    return (np,)


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@app.cell
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@app.cell
def _():
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if __name__ == "__main__":
    app.run()
```

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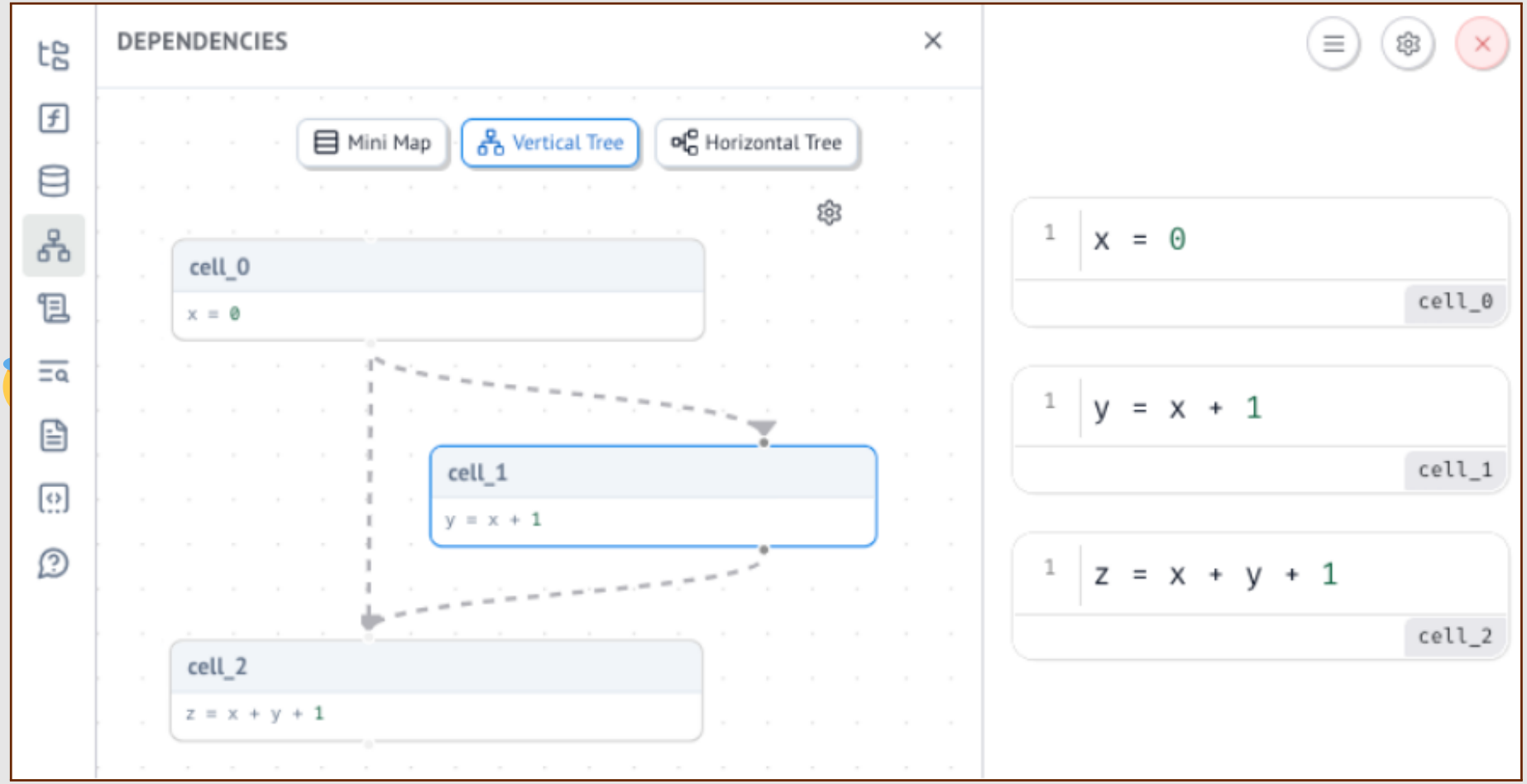

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Marimo notebooks: pure Python

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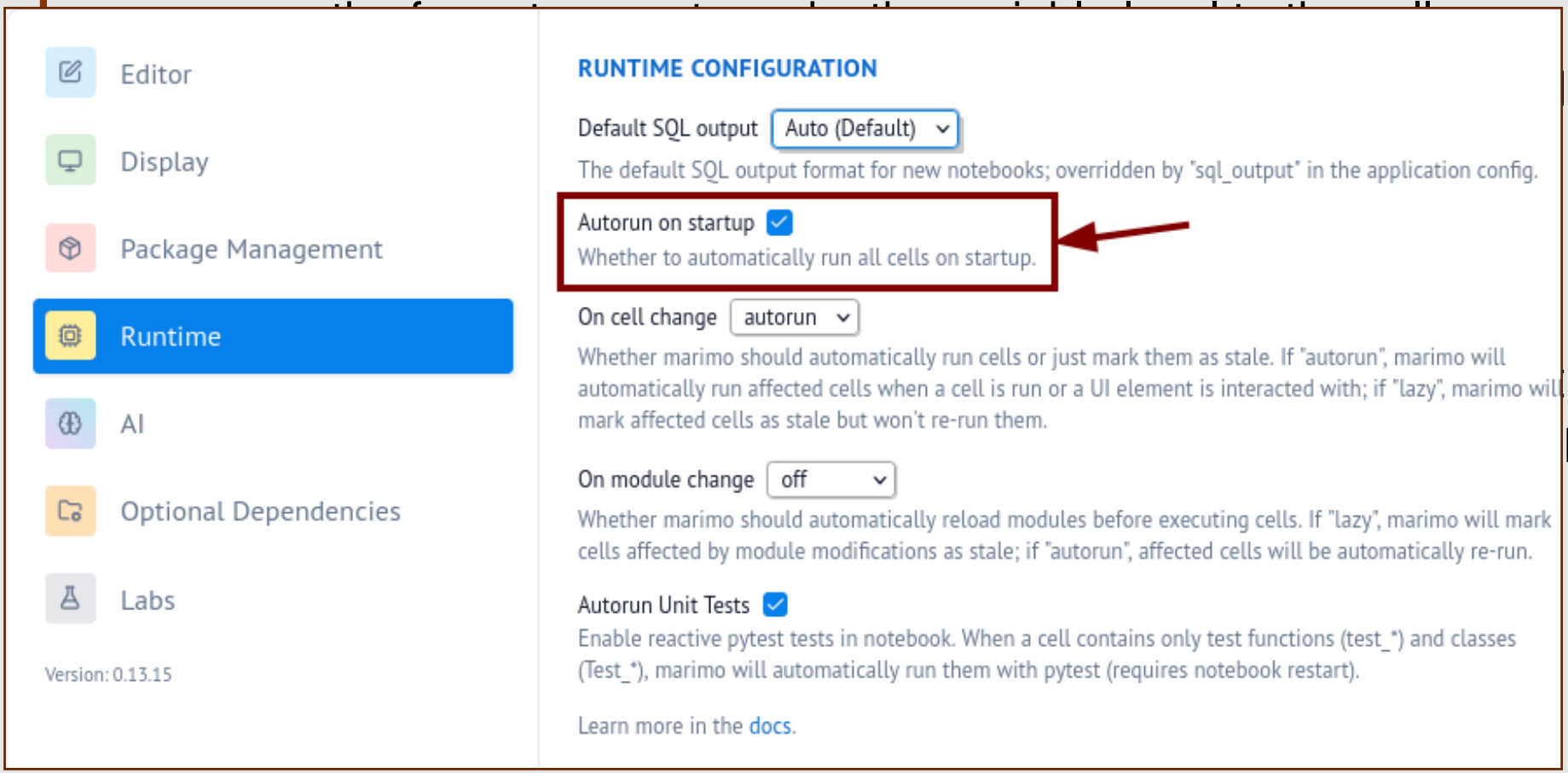

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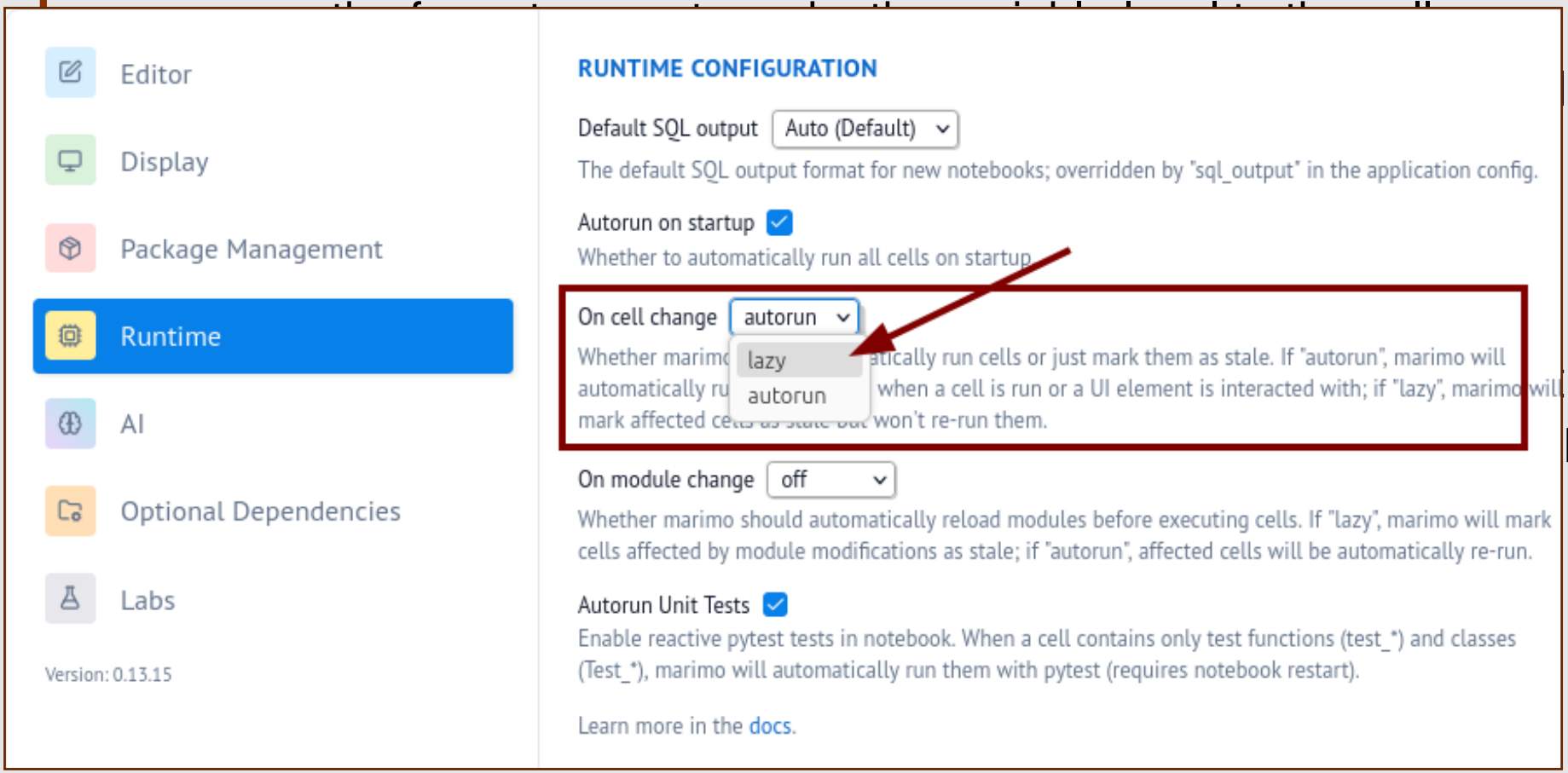

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Marimo installation & run

- marimo installation with uv (see docs.astral.sh/uv/)

```
uv add marimo        # installs the marimo module within the virt. env.
uv run marimo edit   # runs marimo editor within your favorite web browser
```

- marimo installation with pip

```
pip install marimo
marimo edit
```

- marimo installation with conda (universal virtualenv)

```
conda install -c conda-forge marimo
marimo edit
```

Marimo notebooks examples

- `1-simple.py`
- `2-reactive.py`
- `3-interactive.py`
- `4-interactive-plot.py`
- `5-ML.py`
- `5-ML_diskcache.py`

Bibliography I

Jupyter notebooks pros/cons...

Marimo Notebooks

UV